

The Joshi ATS Series under the Bridge-Certificate Standard: A Targeted Substance Audit and an All-Place Rigidity Classification of Effective Normalization Weights

Lukas Geiger

Independent Researcher, Bernau, Germany

ORCID: 0009-0005-7296-1534

October 2026 — Version 0.3 (draft)

Abstract

This ongoing targeted audit separates coordinate bookkeeping, normalization weights, and volume carriers in selected load-bearing steps of Joshi's ATS series. The period-map argument requires consistent raw and normalized logarithmic coordinates and a well-defined action on the period image. Same-field all-place product-formula rigidity forces final effective weights to be constant across places, but permits a global j^2 scaling; its application across L'/L and to the source's freeze remains conditional. A local maximal-order hull inclusion is available under an actual module-hull contract, while the ATS logarithm, Haar-degree, tensor and target-volume bridges remain open. The printed ATS III v4 corollary has absolute-value bars: under a finite nonempty Tate-place sum with $0 < |q_w| < 1$ and positive ℓ, ℓ^* , its left side is nonpositive and its right side positive. This narrow sign inconsistency is separated from ATS IV applications and possible repairs. For global linear quantity families with defined reference sums, rigidity gives at most a y -dependent scalar; classical height and degree follow only with the stated normalization and domain identifications. Measure and log-volume class membership and nonmembership are not established. No conclusion about the truth of abc, the correctness of IUT, or the whole ATS chain is claimed.

Contents

1	Introduction	3
1.1	Mandate and place in the series	3
1.2	Method	3
1.3	Claim boundaries	3
2	The Proof Spine of the Series	4
2.1	The chain	4
2.2	Import triage	5
2.3	Proof text at the load-bearing nodes	5
2.4	What was selected for verification	5
3	Line-Anchored Findings	6
3.1	The normalization mechanism	6
3.2	The period map: ATS II(1/2), Theorem 5.10.1	6
3.3	The forced renormalization: ATS III, Corollary 4.2.2.4	7
3.4	The global period map: ATS III, Theorem 4.6.1	8
3.5	The core of the lower bound: ATS III, Theorem 9.11.1	9
3.6	The translated lower bound: ATS III, Corollary 9.11.1.1	10
4	Typed Gap Catalogue and Verdict Table	11
4.1	Verdict table	11
4.2	Documented reconstructions	11
4.3	A comparative observation	11
5	The Rigidity Classification of the Normalization Weights	11
5.1	Hypothesis labels	11
5.2	The readings of the weight family	12
5.3	Applicability gate: range of places	13
5.4	The classification from Part 6	13
5.5	Quantity-class bridge: what follows and what does not	13
6	Response-Verification Template and Conclusion	14
6.1	The decision tree	14
6.2	Conclusion	15

1 Introduction

1.1 Mandate and place in the series

This series examines the published landscape around the abc conjecture through a fixed auditing instrument. Part 1 [9] localized the disputed step of Mochizuki’s Corollary 3.12 [7, 8]; Parts 2–3 developed the instrument: Part 2 introduced the *bridge contract* criterion [10], and Part 3 its certificate form [11] (the v1-F standard), which fix in advance what a transport argument between two frameworks must deliver before its conclusion may be used; Part 4 [12] applied the certificate to Kirti Joshi’s ATS series as a *form audit*: a pre-specified four-row test (F-1–F-4) of whether the chain states its transport law, its compensation mechanism, its second mode, and its cross-container comparability explicitly enough to be audited at all. The outcome of Part 4 was, deliberately, a statement about auditability, not about substance. A tool supplement analyses the normalization identification separately [14], and a companion landscape paper situates the series among the routes of the wider program [15].

The present Part 5 is a *targeted* substance audit of selected load-bearing internal steps — not a complete verification of the chain. Its mandate, fixed before the work began, is the question left open by Part 4: do the selected load-bearing steps of the published chain carry — that is, do the proofs written in the cited versions establish the statements the chain uses at those sites? IUT imports, the existence base, and steps outside the selected sites are outside the audit.

1.2 Method

The audit followed five phases recorded in a contemporaneous internal proof register before the synthesis stage. Phase 0 mapped the *proof spine*: the minimal chain of results leading from the abc-type inequality back to the existence of the deformation space, with an import triage separating the chain’s own results from statements quoted from the IUT corpus. Phases 1–3 then verified, line-anchored against the exact arXiv versions named in Section 3, the sites the spine prioritizes as heavily load-bearing: the full text of the normalization mechanism, the base package (non-constancy theorem, renormalization corollary, adelic assembly theorem), and the lower-bound core together with its corollary translation. Phase 4 synthesized the findings into the typed findings reported here; the reading question it left open is addressed by the rigidity classification of Part 6, and Section 5 reports that classification under its named hypotheses.

Three disciplines from the earlier parts carry over unchanged. First, *verbatim anchoring*: every finding cites the exact version and, where line-level extracts exist, the line range of the passage it concerns. Second, *typing*: findings are classified as formal bookkeeping (verified or reconstructible), typed gaps (the text does not show what is used — with the missing step named), or documented reconstructions of our own (marked as such, never silently supplied). Third, *claim discipline*: missing-proof findings are not promoted to falsity; the separate printed sign test is stated only under its explicit hypotheses.

1.3 Claim boundaries

This paper reports the current state of an ongoing, targeted audit. The same-field rigidity result is mathematical and conditional on its complete place, field, and quantity-class hypotheses. Most source findings concern missing implications, not counterexamples to a full theorem. Separately, Section 3.6 identifies a narrow inconsistency in a printed sign inequality under explicit finite Tate-place and normalization assumptions. Neither kind of finding decides the truth of abc, IUT, or the whole ATS chain. No new computation, machine-checked proof, or independent full-chain validation is claimed. The audit’s admissible claims may change when a source revision, repair, or counterargument supplies the missing contracts.

The published v0.2 PDFs and the later, hash-bound bilingual source state are distinct baselines. Intermediate work added Part 2 citation, table labels and a response-route table,

language-specific disclosure and revised layout. The present scientific changes use the inventoried coordinate, field, normalization, hull and sign corrections; the exact original v0.2 TeX was not recovered. These provenance limits do not turn the intermediate source into an independently verified copy of the published source.

2 The Proof Spine of the Series

Phase 0 mapped the published chain backwards, from the abc-type conclusion to the statement that deformation spaces of the required kind exist, using plain-text extracts of the four versions listed at the beginning of Section 3. The map is complete *within* those extracts and truncated, not closed, wherever the chain calls on earlier papers of the construction series that were not available for audit — most consequentially at the existence level, which rests on the first of them [2].

2.1 The chain

Table 1 lists the nodes lying immediately on the main chain, in the order the proof text traverses them. The labels $S1, \dots, S22$ are our own bookkeeping; the theorem numbers are the sources’ own.

Table 1: Load-bearing nodes, backwards from the abc-type conclusion. “Proof” counts extract lines between the **Proof.** marker and the end of the argument: an indicator of checkability, not a measurement.

	Result	Rests on	Proof
S1	IV, Thm. 7.2.1 — abc and arithmetic Szpiro over $\mathbb{Q}$	S2	none
S2	IV, Thm. 7.1.1 — Vojta inequality (declared main theorem)	S3, S9, S10	121
S3	IV, Thm. 6.1.1 — First Main Bound	S4 alone	33
S4	IV, Thm. 6.10.1 — Second Main Bound (log-volume sandwich)	S5, S6, S7, S9	55+160
S5	the middle equality of S4 (no theorem number)	“same global normalization”	2
S6	IV, Prop. 6.10.9 / 6.10.10 / 6.10.12 — upper bound	IUT 4, Thm. 1.10, steps (v)–(vii)	2/1/2
S7	III, Cor. 9.11.1.1 — lower bound, translated	S8, by notational translation	none
S8	III, Thm. 9.11.1 — volume lower bound for $\tilde{\Theta}$	S15 and the hull and volume tools	38
S9	IV, Thm. 5.7.1 — existence of initial theta data	standard literature	196
S15	III, Thm. 4.2.2.1 — $\tilde{\Sigma}_{L'}$ stable; j^2 scaling	ATS II, ATS II(1/2), Fargues–Fontaine	15
S16	III, Cor. 4.2.2.4 — forced renormalization	S15 and the “global constraint”	2
S17	III, Thm. 4.6.1 — global period map on $\tilde{\Sigma}_{L'}$	S18, S15, S16	3
S18	II(1/2), Thm. 5.10.1 — product formula as period map	properties of Y_L and arithmeticoids	5
S20	II(1/2), Cor. 4.2.5 — global Frobenius on Y_L	S21	1
S22	II(1/2), Thm. 5.5.2 — existence of arithmeticoids	Scholze; Kedlaya–Temkin; [Joshi 2021a]	36

In one sentence: S1 rests on the Vojta inequality S2, which rests on the First Main Bound S3, which rests entirely on the log-volume sandwich S4; the *lower* half of that sandwich runs through S7 to S8, the *upper* half is taken over unchanged from IUT 4 (S6), and the *middle equality* S5 is what first joins the two halves into a single chain — supported by the normalization apparatus S15 to S18, which sits on the Frobenius construction S20/S21 and the existence level S22. With the auxiliary nodes not shown, the map has 34 rows: 27 numbered nodes and 7 side rows, of which 15 lie immediately on the main chain.

2.2 Import triage

Each node was tagged by provenance: standard published literature; results the series proves for itself; statements quoted from the IUT corpus; and purely definitional passages. Keeping the third tag apart from the first is the structurally important decision of the mapping, because the IUT corpus is not background literature here but the artifact the chain sets out to replace. The ledger of standard literature has 17 entries, 11 of them load-bearing in the sense that a proof of the chain rests on them directly. The ledger of IUT imports has 13 entries, four of which are *unexecuted full imports*, the entire proof consisting of a pointer: “This is the last equation on [Mochizuki, 2021d, Step (v) on Page 658, Proof of Theorem 1.10] and its proof is all of step (v)” (IV:3361–3362), with the parallel formulations for steps (vi) and (vii) at IV:3372 and IV:3390–3391. These three propositions constitute the entire upper half of the sandwich S4.

That fixes the boundary of this audit. Measured by load the IUT-quoted propositions rank first; measured by novelty they rank last, being the source’s computation verbatim. A line-anchored audit there would be an audit of IUT 4, not of the chain under examination, and lies outside the mandate of Section 1.3. They are recorded and set aside, not verified.

2.3 Proof text at the load-bearing nodes

Of the 34 mapped rows, 21 proofs take one of three short forms: no proof block at all (3), a proof consisting of an external citation (6), and a proof consisting of one sentence of the form “clear”, “immediate” or “tautology” (12). Twelve of the 21 sit at nodes lying immediately on the main chain: “The middle equality is a tautology using the fact that the number fields provided by y'_0 and y_0 are identically normalized” (S5, IV:3272–3273); “This is clear from Theorem 4.2.2.1 and the global constraint placed by the requirement that the product formula holds” (S16, III:1613–1614); “The proof is clear” (S20, II(1/2):1318).

The finding is descriptive. A two-line proof can be entirely sound, and several of those listed are shown in Section 3 to carry exactly as written: brevity here indexes *checkability*, which an audit can address, not correctness, which it cannot address in this form. The counts are counts of extract rows, where page numbers stand as free-standing lines and displays break across lines, so every length figure is an indicator and none carries an argument below.

2.4 What was selected for verification

Phase 0 qualitatively prioritized the nodes by load borne, novelty, and brevity of proof text, novelty being the discriminating factor for the reason just given.

The ranking placed the middle equality S5 first: the only node joining the two halves into one chain of inequalities, carrying no theorem number, dispatched in the two lines quoted above, while the same paper states at three places that local comparisons between y_0 and $\varphi(y_0)$ are precisely not available (IV:3208–3211, IV:772–775, IV:816–819). Part 4 of this series treated S5 under the transport-law row of the bridge certificate and resolved its status there — conditionally: the resolution holds on the bound and free readings of the weight family laid open in Section 3.1 and not on the place-selective reading, and Section 5 reports how the readings are classified under named hypotheses. S5 is therefore not re-audited here.

The five remaining ranked nodes follow, preceded by the full-text question Part 4 had left open. **The normalization mechanism** — what “normalized arithmeticoid” operatively means — comes first, because the reading it fixes is presupposed by the three nodes after it. **S18** carries the entire product-formula apparatus on a five-line proof whose justifying sentence states that normalizations do *not* carry over. **S16** asserts in two lines that the j^2 scaling *forces* a renormalization elsewhere, the precondition for S17 and S18 being well defined at all. **S17** is marked by the author himself as “vital to establishing [Mochizuki, 2021c, Corollary 3.12]” and as the type of proposition “claimed without proof” in the works it competes with (III:1838–1840), then proved in three lines. **S8** is the only substantively independent lower bound in the chain,

its decisive reduction declared clear from the construction. **S7** is the node ATS IV actually cites: no proof block, presented as a notational translation of S8.

3 Line-Anchored Findings

All findings below are anchored to four extracts; all four versions are cited in their revisions of 24 February 2025. The line anchors refer to our internal plain-text extracts and are finding aids, not a public reproducibility standard; a provenance manifest (arXiv version, extraction tool, checksums, line-numbering rule) is planned as a series supplement, and until it exists the anchors should be read alongside the quoted wording, which is verbatim.

- **IV** = ATS IV, arXiv:2403.10430v2 [6];
- **III** = ATS III, arXiv:2401.13508v4 [5];
- **II(1/2)** = ATS II(1/2), arXiv:2305.10398v12 [4];
- **II** = ATS II (local prototype), arXiv:2303.01662v3 [3].

An anchor of the form III:6840–6843 refers to lines of the plain-text extract, not to pages of the published document. The extracts are lossy in one respect that constrains what may be asserted: the II(1/2) extract loses Greek letters, so no finding rests on where a λ stands in a formula of that paper, whereas the III and IV extracts preserve sub- and superscripts (control case III:1558) and the ξ/Ξ distinction. The present S7 formula is checked directly against the archived PDF, whose absolute-value bars were lost by the extract; reconstructed formulas are explicitly distinguished from that printed statement. One boundary recurs in Sections 3.5 and 3.6 and is stated once here: Mochizuki’s Corollary 3.12 is not part of the audited corpus, so the claim that a statement of the chain *is* that corollary (III:6880–6881, IV:3197–3198) is not decidable here, and nothing below decides it.

3.1 The normalization mechanism

The normalization coordinate of ATS II($\frac{1}{2}$) §5.3 is the positive exponent converting the untilt absolute value to the fixed standard one. Under the pointwise identity (5.3.3), direct substitution gives the classical degree on $L^\times$ and height on $\mathbb{P}^n(L)$. This statement does not automatically extend to $\mathbb{P}^n(\bar{L})$ or line-bundle heights. The compensation language in §8 and ATS III, Cor. 4.2.2.4, separately requires an identified selection rule and field/place convention.

For a fixed number field L , all of its places, and real x -independent effective weights satisfying the standard product formula for every $x \in L^\times$, Part 6 gives $\beta_{y,v} = c_y$ across places. This leaves a global scalar, possibly dependent on y ; a constant global j^2 extension is admissible. What is excluded is unequal final weights under these hypotheses, not every change of raw local normalization coordinates. Tests on $L^\times$ at places of an extension L' do not by themselves establish the corresponding all-place theorem on $L'^\times$.

The source-bound question remains which rule and domain the operative quantity uses. A pointwise-normalized height on $\mathbb{P}^n(L)$, a free global scalar, a coordinate hyperplane, and a measure carrier are distinct objects. No conclusion about the entire deformation apparatus follows without the bridges in Sections 3.2–3.5.

3.2 The period map: ATS II(1/2), Theorem 5.10.1

The source asserts that $y \mapsto H_y$ is non-constant (ATS II($\frac{1}{2}$), Thm. 5.10.1; extract II($\frac{1}{2}$):2161–2243). Its local variation input is available in ATS II, Thm. 10.20.1(3), but the following coordinate and action identifications must be separated from that input.

Coordinates and bookkeeping. Write raw logarithms as $u_v = \log|x_v|$ and normalized coordinates as $z_v = \alpha_v u_v$. The source's degree is $\deg = \sum_v z_v$, whereas its displayed weighted hyperplane is $H = \ker(\sum_v \alpha_v z_v)$. Thus

$$\ker\left(\sum_v z_v\right) \neq \ker\left(\sum_v \alpha_v z_v\right) \quad \text{in general.}$$

For $\alpha = (2, 1)$ and $z = (1, -1)$ the first sum is 0 and the second is 1. This is a small counterexample to an unqualified kernel identification. It refutes neither the full theorem nor a consistently formulated raw-coordinate reading. A raw-log hyperplane $\ker(\sum_v \alpha_v u_v)$ is a separate consistent contract; it must not be silently identified with the twice-weighted one. The direct-sum support bookkeeping can be retained in its defined domain, but the former blanket assertion that parts (1)–(5) carry is withdrawn.

Translations and the period image. A linear subspace H is invariant under translation by t exactly when $t \in H$. Here this requires $\log_y(x)$ to lie in the actually chosen hyperplane. The product formula for $\sum z_v$ alone does not establish membership in $\ker(\sum \alpha_v z_v)$. An action on arithmeticoids descends through a period map F only if

$$F(y) = F(y') \implies F(gy) = F(gy')$$

for the relevant action elements g . That well-definedness is a separate source obligation, not a consequence of local norm variation.

Non-proportionality and the remaining reconstruction. For a consistently defined weighted kernel, multiplying every coefficient by the same nonzero scalar leaves the hyperplane unchanged. Non-constancy of λ_y therefore does not alone separate hyperplanes; a pair of non-proportional coefficient families is needed. Reconstruction R-1 remains conditional on H-REC: the varied non-archimedean local point can be inserted while the other coordinates are held fixed. ATS II, Thm. 10.20.1(3), supplies the local variation input, not this recombination or the period-image action. The archimedean reach remains unverified. No unconditional S18 closure is claimed.

3.3 The forced renormalization: ATS III, Corollary 4.2.2.4

ATS III, Cor. 4.2.2.4 (extract III:1600–1614), says that the j^2 scaling at $V^{\text{odd,ss}}$ forces a suitable renormalization at the remaining places. This set concerns odd residue characteristic with bad semistable, split-multiplicative reduction (ATS III §3.2(6)), not supersingular reduction. Necessity, localization, existence, and uniqueness are separate claims.

A baseline-relative necessity test (R-2). Suppose S is a nonempty finite set of non-archimedean places of L' , the tested global elements lie in $L^\times$, and the places restrict to the specified usual absolute values on L . Relative to the additional baseline of unchanged normalization coordinates away from S , the scaled product has factor

$$\left(\prod_{w \in S} |x|_w^{\text{std}}\right)^{j^2-1}.$$

Choose a rational prime r under one $w_0 \in S$, so $r \in \mathbb{Q}^\times \subset L^\times$. Then $\log|r|_w \leq 0$ on S and is strictly negative at w_0 . For $j \geq 2$ the displayed factor differs from 1. This is a global element in the correct test domain, not an arbitrary local uniformizer or a principal-ideal generator potentially lying only in L' . Nonemptiness of S and the baseline are additional hypotheses; neither is inferred merely from $\ell \geq 5$.

Same-field rigidity and field descent (R-3). Let b_v be the standard modulus family on one number field K , with the canonical local degree factors included, so that $\sum_v \log b_v(x) = 0$ for every $x \in K^\times$. If $\rho_{y,v} = b_v^{a_{y,v}}$ and $\lambda_{y,v}$ is the raw normalization coordinate, the effective coefficient is $\beta_{y,v} = a_{y,v} \lambda_{y,v}$. Rigidity constrains β , not a and λ separately. For this modulus convention it gives $\beta = c\mathbf{1}$. If one instead uses raw extended absolute values, the conclusion is proportionality to their canonical degree-weight vector, not unweighted constancy. It requires all places of K , including archimedean ones, real x -independent coefficients and the full standard weighted product formula for every $x \in K^\times$. Tests only on $L^\times$ across $w \in V_{L'}$ do not imply constancy of each individual L' -weight. An explicit L'/L descent/aggregation identity or the appropriate full $L'^\times$ formula is needed. The arithmetic transfer is open; the same-field theorem is not disputed.

What a genuine common freeze would imply. Under those same-field identifications and a nonempty freeze set, comparing two solutions at the same j with identical $a_{j,w}$ and frozen $\lambda_{j,w}$ gives identical frozen $\beta_{j,w}$; constancy then gives uniqueness if a solution exists. Nonemptiness is a stated prerequisite, not a verified general assertion about the source. If, in addition, the common reference family really satisfies

$$\rho_{j,w} = \rho_{1,w}^{j^2}, \quad a_{j,w} = j^2 a_{1,w}, \quad \lambda_{j,w} = \lambda_{1,w},$$

at a frozen place, then algebraically

$$c_j = \beta_{j,w} = j^2 \beta_{1,w} = j^2 c_1.$$

This admissible global extension does not violate rigidity. Its source application still requires the common field, norm, place, and freeze identifications; it yields no general verdict of projective emptiness.

The raw-coordinate freeze comparison. To avoid confusing two types of exponents, denote by $A_{j,v}$ the coefficient vector in the fixed raw-coordinate comparison, not the local absolute-value exponent $a_{j,v}$ above. In the inventoried comparison, for $\emptyset \neq S \neq V$ and $j \geq 2$, it has entries $\lambda_{1,v}$ on S and $j^2 \lambda_{1,v}$ outside S , and equals $j^2 \lambda_{\text{can},j,v}$ under that comparison's canonical contract. Consequently

$$H_{\text{Freeze},j} = H_{\text{can},j}, \quad H_{\text{can},j} \text{ need not equal } H_1.$$

This conditional comparison does not certify the source's missing field identification. Nor may a blanket projective-information conclusion be drawn by identifying raw and effective vectors across different fields. The literal finite-place convention of ATS III and its possible archimedean completion remain distinct open source contracts.

3.4 The global period map: ATS III, Theorem 4.6.1

The theorem has seven parts (III:1746–1817) and transports the apparatus of Section 3.2 from a single arithmeticoid over L to an ℓ^* -tuple over L' constrained to Mochizuki's ansatz $\tilde{\Sigma}_{L'}$. Its complete proof: “This is clear from [Joshi, 2023a, Theorem 5.10.1], and the properties of $\tilde{\Sigma}_{L'}$ established so far and detailed in Theorem 4.2.2.1, Corollary 4.2.2.4” (III:1819–1820).

Parts (1)–(3) and (5) cite the componentwise supplier, with separately indexed target spaces (III:1749–1751). Their formal transport is conditional on the consistent coordinate and action contracts of Section 3.2; the former unqualified carry verdict is withdrawn. The tuple extension is asserted in a source remark (II(1/2):2267), rather than proved there. Part (4) explicitly invokes Corollary 4.2.2.4 (III:1791) and inherits the field and normalization obligations of Section 3.3. Part (7) remains outside the passages examined.

The missing restriction step (T17-6). Part (6) asserts that $z \mapsto (H_{y_1}, \dots, H_{y_{\ell^*}})$ is non-constant on $\tilde{\Sigma}_{L'}$, whereas the supplier, Theorem 5.10.1(7), asserts non-constancy on all of Y_L (II(1/2):2228–2233). But $\tilde{\Sigma}_{L'}$ is a proper subset of the corresponding product, cut out by the sharp conditions of Definition 4.2.2 (III:1460–1467), and non-constancy on a superset does not imply non-constancy on a subset. None of the three sources cited in the proof addresses that passage.

Reconstruction R-4: a conditional route. The source supplies stability of $\tilde{\Sigma}_{L'}$ under global Frobenius (Thm. 4.2.2.1(2), III:1536–1539) and the local norm relation $|p|_{K_{y_{n-1}}} = |p|_{K_{y_n}}^{1/p}$ (ATS II, Thm. 10.20.1(3)). Neither statement alone proves non-constancy of the period map. For the proposed route one must fix a common coordinate space and identify the actual hyperplane coefficient vector B_y in those coordinates. One must then prove that a single admissible global move simultaneously realizes the proposed local changes $\lambda_v \mapsto p_v \lambda_v$, that these changes induce the claimed transformation of B_y in the same fixed coordinates, and that B_{φ_y} and B_y are actually non-proportional. Only for nonzero linear forms on that fixed space does this imply $\ker B_{\varphi_y} \neq \ker B_y$. Together with preservation of $\tilde{\Sigma}_{L'}$ and a well-defined period map, this would establish non-constancy on that set. These are H-FROB obligations, not consequences of varying residue characteristics or raw λ coordinates alone. The standard-point and archimedean caveats remain; the required global, coordinate and coefficient identifications are not verified here.

A convention question that decides vacuity. ATS III writes the logarithm and degree maps of parts (1) to (3) without the coefficients λ_v (III:1760, III:1772, III:1781), and its standing convention declares all arithmeticoids normalized. Read literally, the functional is the coordinate sum and H_{y_j} is the same hyperplane for every z , which would leave part (6) without content rather than merely unproved; the statement becomes non-vacuous only under the ATS II(1/2) formulation with the λ -coefficients (II(1/2):2212–2216), which ATS III does not reproduce here. This is a text finding, not an artifact of extraction.

Verdict. *Formal bookkeeping* for parts (1)–(3) and (5) remains conditional on the corrected S18 coordinate and action contracts. *Typed gap* at part (6): the proof as written does not show non-constancy on $\tilde{\Sigma}_{L'}$; the missing step is the restriction from Y_L to that subset. A second, inherited typed gap sits at part (4); a third concerns the valuation convention, under whose literal reading part (6) would have no content, the audited passages not fixing which reading governs. *Documented reconstruction:* R-4 is a conditional route: under H-FROB (Section 5.1) it establishes non-constancy on $\tilde{\Sigma}_{L'}$, using sources the proof does not cite; without an explicit global-to-local compatibility statement, the restriction step remains open both in the source and in our reconstruction.

3.5 The core of the lower bound: ATS III, Theorem 9.11.1

ATS III, Thm. 9.11.1 uses volume, whereas its template Thm. 9.9.1 uses a supremum. Its box obligation must be assessed on the actual carrier, not settled by observing that a single element has measure zero. In particular, the claim that point membership cannot yield a positive-measure hull is withdrawn.

A conditional algebraic hull inclusion. Suppose A is a genuinely split finite local algebra, $\mathcal{O}_A^{\max}$ its specified maximal order, and $\mathcal{O}_{\text{tensor}}$ the tensor order with $\mathcal{O}_{\text{tensor}} \subseteq \mathcal{O}_A^{\max}$. Assume that the actual hull(S) contains S and is stable under multiplication by $\mathcal{O}_A^{\max}$, for example because it is the corresponding maximal-order module hull. Then

$$x \in S \implies x\mathcal{O}_{\text{tensor}} \subseteq x\mathcal{O}_A^{\max} \subseteq \text{hull}(S).$$

The last inclusion uses that module-hull hypothesis. A split algebra and a maximal order alone do not establish it for a merely convex or $\mathbb{Z}_p$ -module hull. A positive full-dimensional box additionally requires the componentwise nonvanishing and the actual local carrier. No silent identification with the ATS target hull or $\tilde{\Theta}$ is made.

Measure normalization and the outstanding volume contract. For a finite extension $E/\mathbb{Q}_p$, normalized additive Haar measure $\mu_E(\mathcal{O}_E) = 1$, and the absolute value $|a| = |N_{E/\mathbb{Q}_p}(a)|_p^{c/[E:\mathbb{Q}_p]}$ with $c > 0$ and $|p| = p^{-c}$, change of variables gives

$$\mu_E(a\mathcal{O}_E) = |a|^{[E:\mathbb{Q}_p]/c} \quad (a \in E^\times).$$

It is not generally $|a|$ under an arbitrary absolute-value convention. The concrete logarithm image, ATS target hull, tensor and Γ balance, degree weights, collation and index layer must still be identified. The abstract algebraic inclusion does not supply that volume theorem.

The finite-sum exponent identity used in the old comparison, $\sum_{j=1}^n j^2/n^2 \leq n$ for $n \geq 1$, remains an elementary arithmetical fact. Its application to a measure lower bound requires the specified Haar, degree, and carrier contracts; the former blanket “exponent arithmetic above the gap is sound” volume verdict is withdrawn. The provenance and transport of the Ξ objects through $\log_{BK}$ also remain separate source obligations (ATS III §§ 9.8–9.11).

3.6 The translated lower bound: ATS III, Corollary 9.11.1.1

The printed statement. The archived ATS III v4 PDF, printed p. 128, Cor. 9.11.1.1, actually has absolute-value bars. With its notation, the displayed formula is

$$-\frac{1}{\ell^*} \left| \text{LogVol}(\tilde{\Theta}_{\text{Mochizuki}}^I) \right| \geq - \sum_{\substack{p, w \in V_p^{\text{odd,ss}} \\ V_p^{\text{odd,ss}} \neq \emptyset}} \log \left| q_w^{1/(2\ell)} \right|_{L'_w}.$$

The summation is over the displayed prime blocks and their indicated places; there are no additional real weights in this printed formula. The old extraction lost the bars. The proposed atomic alternative reading, the claim that the corollary needs bars supplied, and the former exculpation through that alternative reading are explicitly withdrawn.

A narrow sign inconsistency, with hypotheses. Assume $\ell^* > 0$, $\ell > 0$, finite real log-volume, and a defined finite, nonempty Tate-place sum with the usual Tate absolute-value normalization $0 < |q_w|_{L'_w} < 1$. Each $\log |q_w^{1/(2\ell)}|_{L'_w}$ is strictly negative. The left-hand side is nonpositive and the right-hand side is strictly positive. Thus the printed inequality cannot hold under these explicit assumptions. This is a local sign inconsistency of that displayed assertion, not a verdict on the whole ATS chain, IUT, or abc. Empty sums, divergent expressions, or different norm conventions are not silently covered.

What this does not establish. A lower bound $\text{Vol} \geq R$ with $0 < R < 1$ does not imply $\text{Vol} \leq 1$ or $\text{LogVol} \leq 0$; $\text{Vol} = 2$ is a counterexample to that implication. The source’s separate local nonpositivity assertion must not be substituted for a proved volume contract. The criticism of the printed ATS III form, its use in ATS IV, and any corrected lower-bound statement require separate comparisons. A repaired sign convention alone would not close the hull/measure obligations of Section 3.5. H-SIGN below records the actual printed bars and the sign-test assumptions, not an alternative reading.

Table 2: Verdict by operation; open bridges are not whole-chain verdicts.

Operation	Current scope
S18 coordinates/action	direct-sum bookkeeping only in its domain; weighted kernel and period-image action separate
S18 non-proportionality	R-1 conditional on consistent coordinates and H-REC; archimedean reach open
S16 normalization	same-field rigidity valid; L'/L , nonempty freeze and norm identifications open; global j^2 allowed
S8 hull/volume	local inclusion conditional on actual module-hull stability; ATS Haar/tensor/target-volume contract open
S7 printed corollary	absolute-value bars present; narrow sign inconsistency under H-SIGN; IV and repairs separate
S17 restriction	source restriction step open; R-4 conditional on H-FROB

4 Typed Gap Catalogue and Verdict Table

4.1 Verdict table

4.2 Documented reconstructions

R-1 is the non-proportionality route under consistent coordinates and H-REC; R-2 is the baseline-relative necessity test with a global rational prime; R-3 is same-field rigidity, whose source application requires field descent; R-4 is the conditional global-period restriction route under H-FROB. The local hull inclusion is conditional on its actual maximal-order module contract. None closes the complete ATS volume or period chain.

4.3 A comparative observation

Within the comparison corpus of Part 1 [9] and the present audit, both programs contain an unexpanded containment-type transition at a structurally load-bearing site: the program the chain seeks to replace asserts its decisive step with “follows formally” (cf. Part 1), the chain under audit asserts its box containment with “clear from the construction” (Section 3.5). We do not assert identity of the objects, equivalence of the missing implications, or a new audit of IUT here; the observation is a comparison of *assertion form* at the respective critical sites, no more. The difference in form otherwise remains as Part 4 recorded it — the chain under audit states a theorem with a written-out proof and nameable repair obligations. The elementary exponent identity is distinct from its still-open measure application. Neither program is refuted by this observation.

5 The Rigidity Classification of the Normalization Weights

This section applies the rigidity results of Part 6 [13] to the normalization center isolated in Section 3.1. The statements below are implications under explicitly named hypotheses, not pairwise disjoint readings of the source; the hypotheses are defined once here and used consistently.

5.1 Hypothesis labels

H-PLACES. One fixed number field K and its full set of places, including archimedean ones, with an identified standard absolute-value family. An abbreviation is admissible only when the omitted factors are demonstrably retained. Tests on $L^\times$ at L' -places require a separate descent bridge.

H-PF. Real, x -independent final effective weights satisfy the full standard product formula for every $x \in K^\times$, without suppressed place, fiber, or correction terms.

H-5.3.3. The pointwise identity (5.3.3) applies at every specified place and identifies the effective coefficients with $\beta_{y,v} = 1$. Its use on $L^\times$ and $\mathbb{P}^n(L)$ does not establish extension-field or line-bundle transports.

H-EFF. The quantity has a global *linear* representation

$$Q_y = \sum_v \beta_{y,v} q_v, \quad Q_{\text{ref}} = \sum_v q_v,$$

where q_v are y -independent and both sums are defined with compatible index/order conventions and sufficient convergence for the stated scalar factorization. Finite sums or absolutely convergent real sums suffice. No arbitrary nonlinear factorization is included. Identifying Q_{ref} with Q_{cl} is an additional classical bridge.

H-ID. The operative height/degree is the specifically defined quantity on $\mathbb{P}^n(L)$ or $L^\times$, with no uncontrolled additional y -data. Algebraic extensions and line bundles require separate bridges.

H-REC. The required varied local point can be recombined into an admissible arithmetico-coid while the other specified coordinates stay fixed; consistent log/degree/hyperplane coordinates and any period-image action are separate obligations.

H-FROB. The global Frobenius move of R-4 realizes the required local transformations simultaneously and stays in $\tilde{\Sigma}_{L'}$. It identifies the actual nonzero hyperplane forms in a common fixed coordinate space, proves their stated transformation and actual non-proportionality there, and supplies the well-defined period-map contract. Raw normalization-coordinate variation alone does not suffice.

H-SIGN. The literal printed absolute-value formula of Cor. 9.11.1.1, with $\ell, \ell^* > 0$, finite real log-volume and a finite, nonempty usual Tate-place sum with $0 < |q_w|_{L'_w} < 1$, is subjected to the sign test in Section 3.6. This is not an alternative convention.

H-5.3.3 implies H-PF with $\beta = 1$ in the defined same-field standard contract. H-PF alone yields only $\beta = c_y \mathbf{1}$, not y -independence. H-EFF concerns a defined linear quantity family; it is not a field or place identification and does not classify measures automatically.

5.2 The readings of the weight family

The normalization analysis of Section 3.1 identified three readings of the weight family $(\lambda_{y,v})$ in the audited text, and found that the text does not select among them: a *bound* reading, on which the normalization coordinate is determined by the Artin–Whaples standard normalizations [1] (“the normalization coordinate” of § 5.3); a *free* reading, on which the weights are any real family satisfying the weighted product formula; and a *place-selective* reading, on which a rule assigns genuinely place-selective weights while the product formula continues to hold — this last being what the chain’s *use* of the weights requires. These readings are not disjoint classes of solutions: the bound tuple is itself a product-formula solution, and a place-selective solution, if one existed, would be one too. They are alternatives for what the source *means*; which of them the source intends is a question of text interpretation that the mathematics below does not decide. What the mathematics decides is which of these readings are *realizable* by exponent families under the named hypotheses.

5.3 Applicability gate: range of places

The rigidity theorem of Part 6 applies to a full all-place product formula. Before it can be applied to the audited passages, the place-range conflict recorded in Section 3.3 must be confronted: ATS II($\frac{1}{2}$) introduces V_L as all places, archimedean ones included, whereas ATS III §3.2(5), read literally, indexes only non-archimedean valuations and runs its product formula over that range. The audited text does not reconcile these conventions, and power-class equivalence of the local absolute values does not substitute for the missing places. Therefore: (i) if ATS III suppresses but retains the archimedean factors, and the full same-field test domain and other H-PF hypotheses are supplied, the rigidity theorem applies to that completed effective-weight vector; (ii) if ATS III's product formula is literally a non-archimedean-only constraint, the theorem does not apply as stated, and the status of that passage remains open pending a separate theorem or an explicit archimedean completion. In the special case where the finite-place compensation is combined with fixed standard archimedean weights, the combined all-place vector falls under H-PLACES and H-PF, and rigidity then forces the finite weights to a single global value; but this combined formulation must be stated by the source, not inferred from notation. Everything in the remainder of this section is asserted under H-PLACES.

5.4 The classification from Part 6

Part 6 v0.3 [13] supplies same-field rigidity and unchanged finite-fiber redistribution. Under H-PLACES and H-PF, $\beta_{y,v} = c_y$. Under H-EFF, with its actual convergence contract,

$$Q_y = c_y Q_{\text{ref}}.$$

Only with the additional identification $Q_{\text{ref}} = Q_{\text{cl}}$ may this be written as $c_y Q_{\text{cl}}$. Under H-5.3.3 the coefficients are 1; direct substitution yields the classical degree on $L^\times$ and height on $\mathbb{P}^n(L)$. Neither algebraic-closure points nor ATS III line-bundle heights inherit this merely by notation.

Table 3: Rigidity consequences with explicit domains and quantity contracts.

Contract	Admissible conclusion
H-PLACES, H-PF	constant final weights; global j^2 remains admissible
H-EFF as well	$Q_y = c_y Q_{\text{ref}}$; classical identification separate
H-5.3.3 and H-ID	classical degree on $L^\times$, height on $\mathbb{P}^n(L)$
Measures, hulls, log-volume	membership and nonmembership in H-EFF unclassified; no rigidity conclusion

For a stabilized expression on a specified nonempty $L^\times$ family, H-5.3.3 gives the same classical value for every defined constituent only under the stated source normalization; its supremum then has that value. Different domains or added carrier data require separate bridges. A general supremum is not silently treated as a linear H-EFF quantity, nor is a scalar pulled through it without the necessary sign/domain assumptions. The coordinate variation of H_y alone is no proof of metric variation.

5.5 Quantity-class bridge: what follows and what does not

The quantity-class bridge is a domain-sensitive obligation, not a generic transfer of coordinate variation. A degree on $L^\times$ or height on $\mathbb{P}^n(L)$ is classified only by its actual H-EFF/H-ID substitution. A stabilized expression requires its own domain and normalization contract. A hyperplane family alone gives no metric variation. Log-volume, hull, Haar measure and box inclusion have neither membership nor nonmembership in H-EFF established here. Thus Part 6 excludes unequal final place weights under H-PLACES/H-PF; it does not decide the entire height, measure or log-volume chain without the identified quantity/domain bridges. No downstream-node failure is inferred from the classification alone.

Table 4: Quantity-class obligations from weights to operative carriers.

Carrier	Required bridge
$L^\times$ degree / $\mathbb{P}^n(L)$ height	defined linear substitution and classical reference identification
Stabilized expression	specified nonempty domain, normalization and any supremum/scalar conditions
H_y	consistent coordinates and well-defined period action; no metric conclusion by itself
Hull, measure, log-volume	actual carrier, Haar-degree, tensor/ Γ and volume contract; class status unclassified

6 Response-Verification Template and Conclusion

6.1 The decision tree

The findings of this paper concern the published text state at the audit dates given. A future revision of the ATS series, or a reply by its author, is the expected and welcome next event; the following decision tree, fixed here in advance, records the *alternative* routes by which such a document could change the findings. The routes are not cumulative requirements: each branch applies only if that route is chosen, and a legitimate refutation of this audit itself is a further admissible outcome on every branch.

Table 5 states the route logic before the prose checklist.

Table 5: Response routes and what each would need to establish.

Route	What would change	Minimum showing
Misidentification	H-ID no longer attaches the classification to the load-bearing quantity	identify the operative quantity and its additional y -data
Exponent-class repair	the source remains within H-EFF but repairs the missing all-place data	settle H-PLACES, give a selection rule, and track any global scalar c_y
Outside-H-EFF carrier	the rigidity theorem no longer obstructs the chosen carrier	define the measure or log-volume carrier and prove that it carries the needed variation
Lower-bound route	normalization is no longer the decisive step	prove full box containment or supply an alternative lower-bound theorem
Text/citation completion	documentary closures become reconstructible	write the R-1/R-4 closures with sources

1. **Misidentification route.** Show that the operative load-bearing height is not the height of Definition 8.2.1, or that it contains additional y -data — i.e., refute H-ID for the audited passages. The classification of Section 5 then no longer attaches to the load-bearing quantity.
2. **Repair within the exponent class.** If this route is chosen: settle the place range of Section 5.3 (H-PLACES), state an explicit selection rule or a proof of choice-independence for the forced renormalization (the well-definedness obligation of Section 3.3), verify the rule against the one-dimensional solution space at $L = \mathbb{Q}$, and track the effect of a possible global scalar c_y through every used expression.
3. **Carrier outside the effective-weight class.** If the y -dependence is to be carried by a quantity outside H-EFF — for instance at the level of measures or log-volumes — define

that quantity, give its address in the chain, and prove that it carries the required variation; the rigidity theorem does not obstruct this route, and nothing in this paper closes it.

4. **Lower-bound route.** Independently of normalization: either prove the full box containment used in Thm. 9.11.1 (Section 3.5), or supply an alternative lower-bound theorem with a complete derivation.
5. **Text and citation completion.** After any of the above: write the reconstructible closures of Sections 3.2 and 3.4 into the text with their sources. This is a documentation obligation, not a mathematical precondition of the other routes.

6.2 Conclusion

The current targeted audit retains same-field product-formula rigidity and the distinction between non-constancy and non-proportionality. It separates raw and normalized coordinates, the period-image action, field/place descent and actual freeze identifications. Global j^2 scalings are admissible; unequal final weights are excluded only under the complete same-field all-place hypotheses. A conditional algebraic module-hull inclusion is available, while the ATS logarithm, Haar-degree, tensor and target-volume contracts remain open. The actual printed absolute-value corollary has the narrow sign inconsistency stated in Section 3.6; its use in IV and any repaired statement are separate. Linear quantity-class factorization requires defined reference sums and convergence, and classical heights require their stated domains. Measure membership and nonmembership remain unclassified.

This is the ongoing research project's present scope, not an independent verification of the whole ATS chain or a proof/refutation of abc or IUT. The response routes above remain open to a source revision, explicit bridge, or a counterargument to this audit itself. No new computation or machine-checked proof is supplied.

AI Disclosure

AI systems assisted source work, mathematical discussion, drafting, translation and typesetting. Historical adversarial reviews and their limited scopes are recorded in the research files. The present corrective preparation uses the inventoried findings and a narrow original-page check; it introduces no new numerical experiment or machine-checked proof. Independent acceptance of this candidate is not claimed by its preparation.

References

- [1] E. Artin and G. Whaples, *Axiomatic characterization of fields by the product formula for valuations*, Bull. Amer. Math. Soc. **51** (1945), no. 7, 469–492. DOI: 10.1090/S0002-9904-1945-08383-9
- [2] K. Joshi, *Construction of Arithmetic Teichmuller Spaces I*, arXiv:2106.11452v4 [math.AG], 2021 (revised 24 February 2025).
- [3] K. Joshi, *Construction of Arithmetic Teichmuller spaces II: Proof of a local prototype of Mochizuki's Corollary 3.12*, arXiv:2303.01662v3 [math.AG], 2023 (revised 24 February 2025).
- [4] K. Joshi, *Construction of Arithmetic Teichmuller Spaces II₂¹: Deformations of Number Fields*, arXiv:2305.10398v12 [math.AG], 2023 (revised 24 February 2025).

- [5] K. Joshi, *Construction of Arithmetic Teichmüller Spaces III: A ‘Rosetta Stone’ and a proof of Mochizuki’s Corollary 3.12*, arXiv:2401.13508v4 [math.AG], 2024 (revised 24 February 2025).
- [6] K. Joshi, *Construction of Arithmetic Teichmüller Spaces IV: Proof of the abc-conjecture*, arXiv:2403.10430v2 [math.AG], 2024 (revised 24 February 2025).
- [7] S. Mochizuki, *Inter-universal Teichmüller Theory I: Construction of Hodge Theaters*, Publ. Res. Inst. Math. Sci. **57** (2021), no. 1/2, 3–207; *II: Hodge–Arakelov-Theoretic Evaluation*, *ibid.*, 209–401; *III: Canonical Splittings of the Log-Theta-Lattice*, *ibid.*, 403–626; *IV: Log-Volume Computations and Set-Theoretic Foundations*, *ibid.*, 627–723.
- [8] P. Scholze and J. Stix, *Why abc is still a conjecture*, unpublished manuscript, version of 16 July 2018. Available at <https://www.math.uni-bonn.de/people/scholze/WhyABCisStillaConjecture.pdf>
- [9] L. Geiger, *The Gap in IUTchIII, Corollary 3.12: From Attempted Repair to Structural Diagnosis*, Zenodo, 2026. Concept DOI: 10.5281/zenodo.19960781 (latest version v6.3, 12 August 2026, DOI: 10.5281/zenodo.21899747).
- [10] L. Geiger, *The Bridge-Contract Criterion: A Form Audit for Transport Arguments, with an Instantiation in the abc Literature*, Zenodo, 2026. Concept DOI: 10.5281/zenodo.21960476 (version 0.2, 16 August 2026, DOI: 10.5281/zenodo.21960477).
- [11] L. Geiger, *The Bridge Certificate: A Line-Level Audit Instrument for Inter-Theoretic Transfer Arguments, with Four Applications to the abc Literature (DRAFT)*, Zenodo, 2026. Concept DOI: 10.5281/zenodo.21925803 (version 0.2, 16 August 2026, DOI: 10.5281/zenodo.21960958).
- [12] L. Geiger, *The Bridge Certificate Applied to the Joshi ATS Series: A Ledger Audit*, Zenodo, 2026. Concept DOI: 10.5281/zenodo.21951155 (version 0.2, 15 August 2026, DOI: 10.5281/zenodo.21951156).
- [13] L. Geiger, *What the Product Formula Allows: A Rigidity Theorem and an Effective-Weight Test for the ATS Height Subchain*, Zenodo, 2026. Concept DOI: 10.5281/zenodo.21952486 (version 0.3, 2 October 2026, DOI: 10.5281/zenodo.23093243).
- [14] L. Geiger, *Global Normalization in Joshi’s Arithmetic Teichmüller Theory: What the Identification Carries — and What Carries the Theorem*, Zenodo, 2026. Concept DOI: 10.5281/zenodo.21924253 (version 1.1, 13 August 2026, DOI: 10.5281/zenodo.21924254).
- [15] L. Geiger, *The abc Landscape: Obstructions, Failed Routes, and HCT Diagnostics (DRAFT)*, Zenodo, 2026. Concept DOI: 10.5281/zenodo.21916900 (latest version 1.1, 13 August 2026, DOI: 10.5281/zenodo.21924338).